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The solution of Eq. (10) leads to the well-known result¹²⁻¹⁵ of

$$V_0(t) = v_0 \sin \theta = v_0 \operatorname{sech}(\lambda t). \quad (13)$$

Examples of line shapes of the initial pulse of oscillations after the delay from switching on are shown in Fig. 6. These line shapes fit very closely to the hyperbolic-secant function except for the slight departure at the trailing tails.

For the full width at half maximum of the pulse Δt , we have from Eq. (13),

$$\frac{1}{\Delta t} = \frac{3}{2}(\omega_0 a)^{1/2} \frac{v_0}{2.634}. \quad (14)$$

Reciprocals of Δt vs v_0 are shown in Fig. 7. A good proportionality between $1/\Delta t$ and v_0 means that the area of the pulse is kept constant. The value of the constant a calculated from the experimental result is $1.0 \times 10^{-3} \text{ V}^{-2} \text{ s}^{-1}$. This is also consistent with the results obtained previously.

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Equation with the many fathers. The Klein-Gordon equation in 1926

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In this essay I discuss the origin and early development of the first relativistic wave equation, known as the Klein-Gordon equation. In 1926 several physicists, among them Klein, Fock, Schrödinger, and de Broglie, announced this equation as a candidate for a relativistic generalization of the usual Schrödinger equation. In most of the early versions the Klein-Gordon equation was connected with the general theory of relativity. Klein and some other physicists attempted to express quantum mechanics within a five-dimensional unified theory, embracing general relativity as well as electrodynamics. Although this ambitious attempt attracted some interest in 1926, its impact on the mainstream of development in quantum mechanics was virtually nil.

I. THE RELATIVISTIC SCHRÖDINGER EQUATION

"I do not believe that the relativistic equation of 2. order with the many fathers corresponds to reality," wrote Wolfgang Pauli in a letter to Erwin Schrödinger in November 1926.¹ Pauli referred to what is usually known as the Klein-Gordon equation (KG equation in what follows), i.e., the relativistic extension of the Schrödinger equation. Pauli's instinct proved right. Still, the KG equation played a role, though mainly a negative one, in the process which later led to Dirac's wave equation, generally accepted as the correct equation for an electron.²

The usual, nonrelativistic wave equations for an electron are

$$\Delta\psi + 2m/\hbar^2(E - e\phi)\psi = 0 \quad (1)$$

and

$$\left(-\frac{\hbar^2}{2m}\Delta + e\phi\right)\psi = i\hbar\frac{\partial\psi}{\partial t}, \quad (2)$$

as contained in Schrödinger's monumental series of papers published in the spring of 1926. The corresponding KG equations are

$$\hbar^2 c^2 \Delta\psi + [(E - e\phi)^2 - m_0^2 c^4]\psi = 0 \quad (3)$$

and

$$\hbar^2 c^2 \Delta\psi + \hbar^2 \frac{\partial^2 \psi}{\partial t^2} - 2ie\hbar\phi \frac{\partial\psi}{\partial t} + e^2 \left(\phi^2 - \frac{m_0^2 c^4}{e^2}\right)\psi = 0. \quad (4)$$

The equations (1) and (3) are the eigenvalue equations while (2) and (4) are the time dependent equations. In Eq. (3) E signifies the total relativistic energy, including $m_0 c^2$.

The first one to derive Eq. (3), though not to publish it, was probably Schrödinger. As I have shown elsewhere,³ Schrödinger got this equation in mid December 1925, at a time when he had not as yet considered the nonrelativistic equation (1). Schrödinger's original derivation was heavily indebted to de Broglie's formula concerning matter waves. However, no hint of this first derivation exists in the pub-

lished derivations. Schrödinger's original route was essentially as follows. In de Broglie's theory⁴ the phase velocity V and frequency ν of the wave associated with the bound electron in a hydrogen atom were stated as

$$V = E/p = h\nu(1 - \beta^2)^{1/2}/(m_0\beta c)$$

and

$$h\nu = m_0c^2/(1 - \beta^2)^{1/2} - e\phi,$$

where $\beta = v/c$ and v is the mechanical velocity of the electron, equal to the group velocity of its wave. By elimination of β from the two expressions Schrödinger obtained V as a function of ν and ϕ and then inserted V in the standard amplitude wave equation

$$V^2\Delta\psi + 4\pi^2\nu^2\psi = 0. \quad (5)$$

It is easily verified that this leads to Eq. (3) if one takes the further step of substituting ν with E/h .

When Schrödinger published his new wave mechanics in the *Annalen der Physik* the wave equation appeared only in a nonrelativistic form. However, in his first communication he mentioned in words that he had solved also the relativistic eigenvalue equation but that it did not reproduce the fine structure of the hydrogen spectrum correctly.⁵ It was only in his fourth and final communication on wave mechanics, completed in June 1926, that Schrödinger published a general and relativistically satisfying wave equation.⁶ He applied the prescriptions

$$\frac{\partial W}{\partial t} \rightarrow \frac{i\hbar\partial}{\partial t}, \quad \frac{\partial W}{\partial q_k} \rightarrow \frac{i\hbar\partial}{\partial q_k}, \quad (6)$$

to the relativistic Hamilton-Jacobi equation of an electron in an electromagnetic field, written as

$$\left(\frac{\partial W}{\partial t} + e\phi\right)^2 - c^2\sum\left(\frac{\partial W}{\partial q_k} - \frac{e}{c}A_k\right)^2 = m_0^2c^4. \quad (7)$$

W is Hamilton's principal function and A_1, A_2 , and A_3 are the components of the magnetic vector potential. Schrödinger obtained in this way⁷ what he called "the assumed relativistic-magnetic generalization" of Eq. (2), viz.

$$\hbar^2c^2\Delta\psi - \hbar^2\frac{\partial^2\psi}{\partial t^2} - 2ie\hbar\left(\phi\frac{\partial\psi}{\partial t} + c\sum A_k\frac{\partial\psi}{\partial q_k}\right) + e^2\left(\phi^2 - \sum A_k^2 - \frac{m_0^2c^4}{e^2}\right)\psi = 0. \quad (8)$$

Schrödinger did not publish the KG equations in the forms (3) and (4). They are however easily obtained from Eq. (8). In the case of no magnetic field, (4) comes out and if the wave function is required to be periodic in t ,

$$\psi(q,t) = \psi(q)\exp(-i\hbar Et), \quad (9)$$

then (3) comes out.

Schrödinger did not make much use of the relativistic equations which he regarded as less justified than the nonrelativistic versions. When he did use Eq. (4), as in his treatment of the Compton effect,⁸ he associated it with the name of Walther Gordon. But Schrödinger knew of course that the priority belonged to him. When Hermann Weyl, in the second edition of his book *Gruppentheorie und Quantenmechanik* (1931), referred to Eq. (3) as "de Broglie's equation," Schrödinger objected. In a letter to Weyl he wrote⁹:

... it is really somewhat disturbing to read under the headline of de Broglie for instance the operator prescription ($p_i \rightarrow \partial/\partial q_i$) which was, as far as I know, real-

ly communicated for the first time in my note of 18.III.1926, and also the scalar-relativistic equation which I described on the first page of my paper (albeit in words only) and about the solutions to which I have informed in Sec. 3, Chap. 2 of the same paper. I have never protested against this equation being now quite commonly associated with the name of Gordon, because this is very convenient as a distinction. However, for obvious reasons I would rather not see it under the name of de Broglie (also, for example, on p. 188 of your book).

In this connection it is worth mentioning that de Broglie did in fact produce a wave equation of the matter wave of a free electron prior to the advent of wave mechanics.¹⁰ In a paper of February 1925 he arrived at the following equation:

$$c^2\Delta\psi - \frac{\partial^2\psi}{\partial t^2} + 4\pi\nu_0^2\psi = 0.$$

Here ν_0 is the proper frequency of the electron which is related to its mass through $h\nu_0 = m_0c^2$. Apart from a change in sign in the last term, de Broglie's equation is formally identical to Eq. (4) with $\phi = 0$. However, in his paper of 1925 de Broglie interpreted ψ as an electromagnetic wave function associated with the freely moving electron. It was de Broglie's aim to establish a dualistic theory of material particles and phase waves, in contrast to Schrödinger's program which was to substitute wave packets for material particles. De Broglie rejected Schrödinger's view that "material points consist of, or are nothing but, wave systems,"¹¹ and he also objected to Schrödinger's interpretation of ψ as standing waves in an abstract configuration space. In regard of the conceptual differences between the ideas of Schrödinger and de Broglie, de Broglie's equation of February 1925 cannot qualify as an early KG equation in the sense of wave mechanics.

When de Broglie read Schrödinger's first communications on wave mechanics in the spring of 1926 he at once objected to the nonrelativistic foundation of the theory. Many years later he recalled that "the wave equation which he [Schrödinger] attributed to the wave was not relativistic and I was too convinced of the close liaison between the theory of relativity and wave mechanics to be satisfied with a non-relativistic equation."¹² In July 1926, at a time when Schrödinger's relativistic equation (8) had not as yet appeared in print, de Broglie presented complete versions of the KG equations.¹³ De Broglie's method in deriving the eigenvalue equation was similar to the one Schrödinger applied in his process of discovery. Schrödinger's original derivation was of course unknown to de Broglie. Substituting V with $\nu\lambda$ and λ with h/p in Eq. (5) de Broglie got

$$\hbar^2\Delta\psi + p^2\psi = 0.$$

In the relativistic case de Broglie used the relation

$$(E - e\phi)^2 - c^2\sum p_k^2 = m_0^2c^4, \quad (10)$$

which, together with the equation above, led him to Eq. (3). In order to obtain Eq. (4) de Broglie substituted E and p_k with the corresponding operators $i\hbar\partial/\partial t$ and $-i\hbar\partial/\partial q_k$. This substitution is similar to Eq. (6), applied by Schrödinger at about the same time.

II. KLEIN'S APPROACH

The first one to publish a relativistic wave equation was the then 31-year-old Swedish physicist Oskar Klein. As Klein's work has not received much interest from historians of physics I shall deal with it in some detail.

Klein was one of Bohr's first assistants and among his closest collaborators in the early twenties. According to Klein's recollections¹⁴ he began, in 1921–1922, to think about quantum phenomena as the result of some deep-rooted but as yet unknown connection between particles, waves, and quanta. The direction of Klein's early thoughts was not unlike that which guided de Broglie at about the same time. In 1923 Klein became convinced that the motion of atomic electrons could somehow be represented by self-interfering waves; he believed that the motion would be governed by Hamilton's analogy between geometric optics and particle mechanics. At the same time he began to think of quantization of energy states as being eigen solutions to vibration phenomena and he sought for a wave equation in terms of a wave interpretation of the Hamilton–Jacobi equation. Klein's early ideas remained obscure and unfruitful for some years. Unfortunately he complicated things by sticking to wave equations with nonlinear terms and was soon led to mathematics he was unable to cope with. He was, furthermore, handicapped by the lack of scientific response which to some extent was due to his natural shyness. Until 1926 Klein's preliminary wave theory of atoms remained a rather private affair. Commenting on his early ideas, Klein remarked in 1963 that “I had that in the back of my head when I was doing much earlier work, but I feared it was too fantastic to be anything, so I mentioned very little about it. You know, Bohr was also very busy, so I was always afraid of disturbing him. I was thinking very much about it in 1922 in Göttingen, and when I came back from Göttingen ... I mentioned this to him [Bohr], but I saw from his reply that he didn't get what I meant. Then I thought that I had better not disturb him any further.”¹⁵

From September 1923 to June 1925 Klein was occupied as a research assistant at the University of Michigan where he worked under W. Colby and H. M. Randall. During his stay in Michigan Klein was led to speculate that his ideas of a wave theory of atoms might be connected with a unification of Maxwellian electrodynamics and Einsteinian general relativity. He sought to work out a theory based on this idea but at that time it was in vain. His speculations led him to the idea of expressing the quantum conditions in a five-dimensional unified theory of relativity. According to an interview, undertaken in 1963 by John Heilbron and Leon Rosenfeld, the idea emerged as follows:

I think the main thing was that I wanted to have waves with a definite velocity. Then it occurred to me that if you have a particle moving in 3 dimensional space with constant velocity, the projection of that particle on a plane will have different velocities, although the velocity of propagation is given. So that was the first. I got further in the late autumn '24. Then, as far as I remember, I gave a lecture course there on electromagnetic theory, and also on the motion of particles in electric fields. Then I saw that one could write—I suppose that was known, but I don't think that I knew it—the Hamilton–Jacobi equation also for the relativistic motion of an electron in an electromagnetic field in quite a nice way. You had a quadratic form first,

and then you had a linear form with the vector potentials in the momenta, or the momenta would be then the derivatives of the Hamilton–Jacobi function with respect to the x 's. I knew a little of general relativity theory, although not very much. Then I saw that if you wrote the corresponding equation for the motion of a particle according to general relativity, then you would have a square form: (ds/dx_i) times (ds/dx_k) and multiplied by this g^{ik} . Then it occurred to me that these electromagnetic terms could be written in the same way if you assumed that you had a momentum component in the fifth dimension which was constant. That would appear as a constant and then you would have the potentials which would then correspond to the g^{i0} . Then it was clear that the momentum in that direction would be simply connected with the charge of the particles.¹⁶

When Klein returned to Scandinavia in the early summer of 1925 he realized that the nonlinear approach was a failure and considered the simple linear wave equation. Klein later claimed that in the summer of 1925 he had a generalized wave equation which contained Schrödinger's later equations as special cases. “I had it [the KG equation] before he [Schrödinger] began the thing at all, but I hadn't published it. ... I had the general wave equation, I had that quite a time earlier [than Schrödinger].”¹⁷ Klein's claim was made almost 40 years after the events in question. Accordingly the historian of science cannot accept it at its face value. In the absence of primary evidence, such as letters or notebooks,¹⁸ the question of Klein's priority cannot be settled.

While in Michigan Klein was unaware of de Broglie's theory but in the summer of 1925 Bohr supplied him with a copy of de Broglie's thesis.¹⁹ Klein ran through it and noticed that some of de Broglie's ideas were similar to his own. But he was not particularly impressed and did not make a thorough study of de Broglie's theory such as did Schrödinger at about the same time. Unhappily interrupted by a long period of illness in the autumn of 1925 Klein continued his lonesome attempt to formulate a five-dimensional theory of relativity and quanta. As a test case of his five-dimensional theory Klein wanted to find the eigenfrequencies of the harmonic oscillator. This involved him in a series of endless calculations and frustrations, partly caused by his lack of mathematical knowledge (for example, he did not know about Hermite polynomials at the time).

Klein was only able to present an acceptable version of his theory in the spring of 1926, at a time when Schrödinger's first paper on wave mechanics had already appeared. Eventually he published his work on a unification of quantum theory and five-dimensional relativity in *Zeitschrift für Physik*.²⁰ In this paper, submitted at the end of April, the KG equations appear in print for the first time. However, the relativistic wave equation was only of minor importance to Klein who considered the merit of his paper to lie in its five-dimensional theory:

I thought it was quite self-evident as soon as one thought of such a wave equation in relativistic mechanics, which was a known thing. I started the whole thing in relativistic mechanics because I had this five-dimensional approach. I never thought that that was any important thing, just to have the relativistic scalar equation after Schrödinger's equation. But I thought

there was some other things which were important, for instance, that it was time dependent, which, as Heisenberg pointed out to me, could give important applications which one couldn't have from Schrödinger's first work. Of course, Schrödinger very soon got to that himself, so I think he had published that before my paper came out, but not before I had written it.²¹

Klein's idea of a unified theory of gravitational and electromagnetic phenomena in terms of a five-dimensional generalization of Einstein's theory of relativity had already been suggested by the German mathematician Theodor Kaluza in 1921 (see also below). Kaluza did not consider quantum theory. When Klein worked out his theory of a five-dimensional world he was unaware of Kaluza's earlier theory. It was only when Klein returned to Copenhagen from Sweden in March 1926 that Pauli, when reading Klein's manuscript, made him aware of Kaluza's work. Pauli was an authority on relativity and familiar with the entire literature.²² Naturally Klein was disappointed to discover that major parts of his theory had already been treated by Kaluza and by Schrödinger. He had to change his manuscript materially and included references to the works of Schrödinger and Kaluza. At the end of April he submitted his revised manuscript, in "a state of mental resignation."²³

In Klein's theory the motion of electrically charged particles was thought to take place along geodesic lines in a five-dimensional Riemannian space with the line element

$$ds^2 = g_{ik} dx^i dx^k + (dx^0 + \beta A_k dx^k)^2,$$

where the indices i and k run from 1 to 4. The first term is the line element of the ordinary theory of relativity while x^0 is a new fifth coordinate. A_k are the components of the electromagnetic four-potential and β is a constant. As to the physical significance of the fifth coordinate Klein thought that x^0 was a nonobservable quantity; but he emphasized that observability is not a valid criterion of physical meaningfulness and that x^0 was not superfluous.²⁴ Klein suggested that x^0 might be considered as conjugate to the electrical charge so that the "momentum," corresponding to the fifth coordinate, was given by

$$p_0 = e/\beta c.$$

He hoped that this idea would enable him to explain the atomicity of electricity and also account for the basic building blocks of matter, the electron and the proton.

Klein constructed the equation of motion in his five-dimensional world as a ray equation by means of Hamilton's optical-mechanical analogy. He imagined "the observed motion as a sort of projection on the time interval of a wave progression which takes place in a space of five dimensions."²⁵ In the simple electrostatic case, as in the hydrogen atom where gravitational interactions may be neglected, Klein found the following equation of motion:

$$c^2 \Delta U - \frac{\partial^2 U}{\partial t^2} - 2e\phi \frac{\partial^2 U}{\partial t \partial x_0} + (m_0 c^4 - e^2 \phi^2) \frac{\partial^2 U}{\partial x_0^2} = 0. \quad (11)$$

The function $U = U(x_0, x_1, x_2, x_3, ict)$ is a wave function associated with the motion of the electron in five-dimensional space. In deducing Eq. (11) Klein did not make use of Planck's constant or other results of quantum theory. He introduced quantum theory by assuming U to be a periodic

cal function given by

$$U = \psi(x_1, x_2, x_3) \exp[i(2\pi\nu t - x_0/\hbar)].$$

If U is now inserted in Eq. (11) and $(m_0 c^2 + E)$ is substituted for $\hbar\nu$ one obtains Eq. (3) which was stated in this form by Klein. The general, i.e., time-dependent KG equation is contained in Eq. (11). If one replaces U with $\psi \exp(-ix_0/\hbar)$, Eq. (4) comes out.

Klein aimed primarily to establish a unitary theory of electromagnetism, gravitation, and quanta. He did not attach much importance to the four-dimensional wave equation as a separate item and did not explore its physical significance. For example, he did not attempt to find its energy eigenvalues.

III. THE FIVE-DIMENSIONAL THEORY

The five-dimensional theory of relativity, as worked out by Kaluza and Klein, aroused a certain interest in the twenties. In 1926 the theory was investigated by the Russian physicist H. Mandel who came independently to it; just as Klein, Mandel was unaware of Kaluza's work.²⁶ Einstein, who in 1919 had encouraged Kaluza to pursue his idea of a fifth dimension, also contributed to the subject.²⁷ Einstein discussed the five-dimensional theory in two addresses to the Prussian Academy of Science but without mentioning its possible connection to quantum theory. At the end of his second paper he added the following note: "Herr Mandel points out to me that the results communicated by me are not new. The entire content is found in the paper by O. Klein."²⁸ This note, written in February 1927, might suggest that Einstein did not know about Klein's paper until Mandel made him aware of it. That was, however, not the case. In a letter to Ehrenfest of 3 September 1926, Einstein wrote that "Klein's paper is beautiful and impressive."²⁹ Apparently Einstein did not read Klein's paper very thoroughly.

Klein's attempt to incorporate quantum theory in five-dimensional relativity was rather widely discussed during 1926. It was, for instance, taken up by Fritz London who proposed that the fifth dimension might be related to the newly discovered spin of the electron.³⁰ According to London the quantity $m_0 c^2$ might be interpreted as a rotational spin energy of the electron. He proposed that the angular momentum of the electron was the canonical conjugate to x_0 , i.e.,

$$d = -i\hbar \frac{\partial}{\partial x_0}.$$

London did not pursue this idea or other aspects of Klein's theory. He was more inclined to think that a unification of wave mechanics and general relativity should be based on Weyl's unitary theory of 1919, a theory which most physicists found untenable at the time.³¹

When Klein presented his theory to his colleagues at the Institute of Theoretical Physics in Copenhagen he did not receive favorable responses from either Bohr or Heisenberg. ("I think none of them were really interested; I think they were rather skeptical."³²) The British physicist L. Thomas, who was then in Copenhagen, took, however, interest in Klein's theory and gave a talk about it in Leiden on his way back to Cambridge. As a result, Klein was invited to Leiden by Lorentz and Ehrenfest. Klein lectured on his new theory and Ehrenfest and Uhlenbeck got pretty enthusiastic about it. As Uhlenbeck recalled in a conversation

with Abraham Pais: "I remember that in the summer of 1926, when Oskar Klein told us of his ideas which would not only unify the Maxwell with the Einstein equations but also bring in the quantum theory, I felt a kind of ecstasy! Now one understands the world!"³³ In the early summer of 1926 Klein cooperated with Ehrenfest and Uhlenbeck on various problems of wave mechanics. Among other things they investigated dispersion and Zeeman effect by means of the nonrelativistic limit of Klein's general wave equation. A manuscript for an article was prepared but Ehrenfest, whose interest in the five-dimensional theory had soon cooled, did not want to join Klein in publishing. Accordingly the paper was never submitted for publication.³⁴ The impact of Klein's visit resulted in a paper by Ehrenfest and Uhlenbeck, written in August 1926.³⁵ They pointed out that Klein had come to the idea of a wave mechanics independently of Schrödinger and de Broglie, the main difference being that Klein's phase waves U propagate without dispersion.

Another five-dimensional wave mechanics was proposed by the Leningrad physicist Vladimir Fock in July 1926.³⁶ He came to his theory independently of Klein whose article reached Leningrad when Fock's paper was in press. Fock did not base his work on Kaluza's theory but on Mandel's work which he knew from its manuscript. Fock adopted a five-dimensional line element similar to Klein's and also obtained a general five-dimensional wave equation similar to that found by Klein.

Still another five-dimensional wave theory was worked out by Leon Rosenfeld in connection with his doctoral dissertation at the Free University of Brussels.³⁷ Rosenfeld knew about Kaluza's theory but began his work before Klein's paper came out.³⁸ Rosenfeld's series of articles was much influenced by Th. de Donder, his supervisor at Brussels, who had himself proposed a unification of general relativity and wave mechanics (see below). During the preparation of his dissertation Rosenfeld spent part of his time in Paris where he worked under de Broglie. According to Rosenfeld it was he who called de Broglie's attention to the five-dimensional formulation. De Broglie got very interested and wrote a paper in which he gave his own version of Klein's theory.³⁹ Rosenfeld told as follows:

I think de Broglie wanted to avoid the configuration space of Schrödinger because he said it wasn't sufficiently physical; he thought of the waves as something real, and therefore, didn't like the waves in configuration space. Then he thought that perhaps by introducing such a general field, the gravitational field, and especially the five-dimensional formalism which included all the electromagnetism, that you could get closed equations that included the interactions.⁴⁰

The wave equations obtained by de Broglie were similar to those already published by Klein and contained no new physics. Although de Broglie's paper gave full credit to Klein, it caused a minor controversy with the Swedish physicist. Hendrik Kramers, then at Bohr's institute in Copenhagen, had written a paper about static gravitational fields back in 1922.⁴¹ He gave a copy to Klein who at that time hardly knew anything about general relativity and accordingly did not understand it. Later, when Klein prepared his paper for publication, Kramers pointed out that his old paper contained ideas which had some similarity to those developed by Klein. Although Klein had made no use at all of Kramers' paper, he was persuaded by Bohr and

Kramers to quote it in his article where it appeared together with references to Kaluza, de Broglie, and Schrödinger. When de Broglie wrote his paper on five-dimensional relativity wave mechanics he credited Kaluza and Kramers as the founders of five-dimensional theory and Klein as the one who had extended it to cover quantum theory. This was not an unnatural conclusion in regard of the references in Klein's paper. Klein felt however that his own work was not properly credited and that "de Broglie quoted Kramers for the whole thing."⁴² Accordingly he protested against what he felt was unfair crediting and pointed out that Kramers' work was not about five-dimensional relativity at all.⁴³ The insulted Klein received "a nice letter from de Broglie where he said that he was actually a little disgusted by the whole thing."⁴⁴ In a public reply de Broglie said that he had no intention to belittle Klein's contribution which he liked very much. As to the quotations he admitted to have failed:

I only knew the theory of the five-dimensional universe from the paper of M. Klein and it is from that paper that I took my references. I was led to believe that M. Kramers had made a personal contribution to the construction of this theory. I freely admit that as to this point I was mistaken and that only M. Kaluza should be attributed what I attributed to M. Kaluza and M. Kramers.⁴⁵

This small episode is interesting from the point of view of sociology of science. Citations are usually taken to imply that the quoted papers have actually been influential to the scientist who cites them. In Klein's case the references to Kaluza and Kramers were courtesy citations; Klein had read the papers of Kaluza and Kramers but they were of no importance to his own paper. De Broglie quoted Kramers and Kaluza without even having seen their papers. He did not, on the other hand, refer to the work of Fock which he most probably knew and had benefitted from. Evidently citations should be handled with care by the historian or sociologist of science. If one does not distinguish between material and nonmaterial citations, the results to be drawn from citation studies are likely to be distorted.

Not all physicists considered the five-dimensional quantum theory of Klein and others a sound avenue. In Leningrad George Gamow and Dmitri Iwanenko took up the five-dimensional theory of Klein and Fock.⁴⁶ But Iwanenko seems quickly to have lost confidence in the theory. In a subsequent paper, written jointly with Lev Landau, he stressed that "the Klein-Fock equation," that is our Eq. (8), could easily be derived "without the somewhat artificial introduction of the fifth coordinate."⁴⁷ The method applied by Iwanenko and Landau was only slightly different from that applied by Schrödinger.

Pauli, with whom Klein had discussed his work in Copenhagen before publication, was even more critical than usual. In a letter to Bohr he called Klein "a great champion of confusion."⁴⁸ In another letter to Bohr Pauli wrote as follows: "I should however congratulate you on having made Klein only touch on the fifth dimension in the last section. It would, however, have been even better if it had been totally omitted, but after all it is something that you have made him leave it at that."⁴⁹ The article which Pauli commented on in his letter to Bohr was submitted at the end of 1926. It contained rederivations of the full KG equations as well as expressions for the corresponding charge and current densities. It was only at the end of the extensive

paper that Klein restated his five-dimensional theory in which he still had great confidence. In the interview of 1963 Klein recalled Pauli's critical attitude:

I think he thought that it was dishonest that I believed in the five dimension theory and still wrote a great part of the paper just to show how the experimental quantities could come out in a four dimensional way. But I didn't think there was anything dishonest about it. One could say that when one knew, and saw through the thing, that it was not connected because one couldn't carry through the five-dimension view. But at that time I still hoped that one could and that the four-dimensional treatment was a more provisional thing.⁵⁰

The attempts to express quantum mechanics in a world of five dimensions continued for a couple of years. The theory was taken up by mathematical physicists in the U.S.A. and France but resulted in very little of physical interest.⁵¹ As a result of Dirac's successful theory of 1928 the interest in five-dimensional relativistic quantum mechanics largely disappeared. Klein lost his faith a little before Dirac's paper appeared. In an article on five-dimensional relativity theory, published in 1928, he stated in a footnote that "[I] no longer consider it possible to introduce a fifth dimension as a means of accounting for the deviations from the space-time description of classical theory which the quantum theory requires."⁵² According to Klein's memory the advent of Dirac's theory marked the death of the fifth dimension so far as quantum theory was concerned: "At that time I thought there were no reasons for this five-dimensional theory, and I gave it up. I remember that in the spring of '28 after I had been at Cambridge, Pauli was in Copenhagen and we had him to dinner. We had a bottle of wine and we drank to the death of the fifth dimension."⁵³

The announced death of the fifth dimension was a little premature. Klein resumed his interest in the five-dimensional formalism at some later occasions⁵⁴ and also a few other physicists continued to work on the subject.

IV. PAULI'S UNPUBLISHED DERIVATION

After Schrödinger had shown the way in his first publication on wave mechanics, the derivation of a relativistically covariant version of the wave equation was a rather straightforward matter. This is illustrated by the ease with which Pauli got the KG equation in April 1926.

Pauli knew about Schrödinger's first paper early in February. In contrast to some of his German colleagues, in particular Heisenberg, his first reaction was that Schrödinger's theory was "not completely mad."⁵⁵ He immediately began to think about the new wave theory of atoms and its relationship to the matrix mechanics of the Göttingen physicists. In the beginning of April, on one of his travels to Copenhagen, Pauli figured out how to prove the expected equivalence between matrix and wave mechanics.⁵⁶ After having discussed the matter with the Copenhagen physicists, Pauli wrote a letter to Jordan in which he stated his proof in detail.⁵⁷

In the present context the interesting thing about Pauli's letter is that he started with establishing the relativistic wave equation, although in the equivalence proof he used only the nonrelativistic approximation. Pauli's method was similar to the one published by Schrödinger three

months later (see above). Pauli arrived at the energy equation (3) but did not state the time dependent equation (4). The message of Pauli's letter circulated among many physicists who were thus made acquainted with the equivalence before Schrödinger's proof appeared in print. For example, Heisenberg reported the main content of Pauli's letter to Dirac in May.⁵⁸ In this letter Heisenberg also gave the relativistic wave equation in the form written by Pauli, viz.

$$E^2 \Delta \psi - [(E - E_{pot})^2 - m_0^2 c^4] \frac{\partial^2 \psi}{\partial t^2} = 0.$$

This is the same as Eq. (3) if one applies the substitution (9).

Pauli soon reached the conclusion that it would be impossible to establish the required equivalence between matrix and wave mechanics if the latter was based on the KG equation. This insight was a major reason to Pauli's lack of confidence in "the equation with the many fathers." Pauli expressed his point of view in a letter to Gregor Wentzel of July 5:

The differential equation

$$\Delta \psi - \frac{1}{K^2 c^2} (E - E_{pot})^2 \psi + \frac{m_0^2 c^2}{K^2} = 0 \quad (1)$$

($K = h/2\pi i$, E = energy including $m_0 c^2$) has a very uncomfortable property. The fact is that it is not self-adjoint, the associated eigenfunctions ψ_n not being orthogonal. The term $(2/K^2 c^2) E \cdot E_{pot} \psi$, which comes from squaring $(E - E_{pot})^2$, is responsible for this as will be seen immediately when forming $\psi_n \Delta \psi_m - \psi_m \Delta \psi_n$. However, if one wants to keep the connection with the matrices

$$q_{nm} = \int q \psi_n \psi_m dV; \quad p_{nm} = K \int \psi_n \frac{\partial \psi_m}{\partial q} dV, \quad (2)$$

the eigenfunctions must be orthogonal. Therefore I have completely lost my faith in the differential equation (1)!⁵⁹

Pauli felt that the spinning electron had to be taken into account in a relativistic quantum theory. But in spite of many attempts he was unable to formulate a satisfactory relativistic spin theory. The problem was only solved with Dirac's theory.⁶⁰

Much of Pauli's concern with the KG equation took place in collaboration with the Hungarian physicist Johann Kudar who came to Hamburg in October. When still in Budapest Kudar published on the KG equations, after Klein, Schrödinger, Fock, and de Broglie had published their papers.⁶¹ Kudar showed that Schrödinger's first method to obtain the wave equation, i.e., by means of a variational principle, could be carried over into the relativistic domain and would then result in Eq. (3). Essentially the same thing had been shown by Fock in an earlier paper (see below).

As an assistant to Pauli, Kudar was induced to investigate the KG equation in the light of the general formalism of quantum mechanics. He reported about his and Pauli's results in letters to Schrödinger and Dirac. He supplied Schrödinger with Pauli's proof that the solutions of the KG equation are not orthogonal functions.⁶² He ended his letter:

Although the paternity of the relativistic wave equation is a multivalued function (its birth was announced by several authors) and the fathers of this differential

equation would not at all suppress their happiness (cf. Klein, Fock, Kudar) Pauli will, in a postscript to the present letter, express his opinion, as this four-dimensional problem child seems to carry *τα στίγματα θανάτου* [the stigmatization of death] within it.⁶³

V. FOCK'S WORK

Vladimir Fock submitted his first paper on relativistic wave mechanics in the beginning of June.⁶⁴ At that time Klein's work was unknown to him and only the first two of Schrödinger's communications had reached Leningrad. Fock based his derivation of the wave equation on a variational principle inspired by Schrödinger's method in his first communication. Schrödinger's method was to write down the Hamilton–Jacobi equation of motion and replace the momenta with the quantities $\hbar/\psi \partial\psi/\partial q_k$; in this way he got an equation quadratic in ψ and $\partial\psi/\partial q_k$.⁶⁵ In order to obtain the wave equation Schrödinger required that

$$\delta \int \left[H \left(q_k, \hbar/\psi \frac{\partial\psi}{\partial q_k} \right) - E \right] dq_1 dq_2 dq_3 = 0.$$

Fock applied the substitutions

$$\frac{\partial W}{\partial t} = -E \frac{\partial\psi}{\partial t} / \frac{\partial\psi}{\partial t},$$

and

$$\frac{\partial W}{\partial q_k} = -E \frac{\partial\psi}{\partial q_k} / \frac{\partial\psi}{\partial t},$$

where W is Hamilton's principal function and E the energy. In order to get the wave equation for an electron in an electrostatic field, he proceeded as follows. The Hamilton–Jacobi equation is Eq. (7) with $A_k = 0$. With the mentioned substitutions it reads

$$\left(-\frac{E}{c} \frac{\partial\psi}{\partial t} / \frac{\partial\psi}{\partial t} + \frac{e\phi}{c} \right)^2 - E \sum \left(\frac{\partial\psi}{\partial q_k} / \frac{\partial\psi}{\partial t} \right)^2 - m_0^2 c^2 = 0.$$

In order to transform this expression into a quantity Q which is quadratic in $\partial\psi/\partial q$ and $\partial\psi/\partial t$, it is multiplied with $(\partial\psi/\partial t)^2$:

$$Q = [(-E + e\phi)^2 - m_0^2 c^4] \left(\frac{\partial\psi}{\partial t} \right)^2 - c^2 E^2 (\nabla\psi)^2 = 0.$$

Then the variation

$$\delta \int Q dq_1 dq_2 dq_3 dt = 0$$

is applied. When worked out by means of the Euler–Lagrange conditions this leads to the equation

$$\Delta\psi - 1/E^2 \left[2mE + \frac{E^2}{c^2} - 2 \left(m + \frac{E}{c^2} \right) e\phi + e^2 \phi^2 \right] \frac{\partial^2 \psi}{\partial t^2} = 0.$$

Fock considered periodic wave functions given by Eq. (9). Then the energy eigenvalue equation comes out in the form

$$\Delta\psi + 1/\hbar^2 [2mE + E^2/c^2 - 2(m + E/c^2)e\phi + e^2 \phi^2] \psi = 0.$$

Since $E = -\partial W/\partial t$ denotes the classical energy it can be replaced with $(E - m_0 c^2)$ where E is now the total relativistic

energy. In this way the above equation reduces to Eq. (3).

Apart from applying his method to the electrostatic case, Fock also derived and discussed the wave equations for an atom in a magnetic field and an external electric field (Zeeman and Stark effect). In the early days of 1926 Schrödinger applied his variational procedure also to the relativistic case. However, Schrödinger's relativistic calculations were never published.⁶⁶ He obtained the following variation, as stated in his notebook:

$$\delta \int \left((\nabla\psi)^2 - \frac{1}{\hbar^2 c^2} [(E - e\phi)^2 + m_0^2 c^4] \psi^2 \right) \times dq_1 dq_2 dq_3 = 0.$$

It is seen that this equation is included in Fock's expression for Q . If one applies Eq. (9) Fock's variation transforms into Schrödinger's unpublished variation.

Fock was the first one who solved the energy equation (3) for the hydrogen atom, in which case $\phi = -e/r$. Schrödinger had done so half a year earlier but in his papers he only referred to these calculations in words. The case of the hydrogen atom was crucial to Schrödinger who was deeply worried to discover that the relativistic equation does not reproduce the fine structure correctly.⁶⁷ Fock adopted the method that Schrödinger had developed in his first communication and there applied to the nonrelativistic approximation (1). Fock reached of course the same result as stated by Schrödinger, i.e., Sommerfeld's fine structure formula with wrong (half-integral) quantum numbers. Fock's result for the energy values of the hydrogen atom was

$$E = m_0 c^2 \left[\left(1 + \alpha^2 / \left((p - \frac{1}{2}) + [(n + \frac{1}{2})^2 - \alpha^2]^{1/2} \right)^2 \right)^{-1/2} - 1 \right],$$

where p and n are integral quantum numbers corresponding to the principal and azimuthal quantum number appearing in Sommerfeld's formula. Shortly after Fock's work was the hydrogen fine structure calculated also by Carl Eckart at Caltech.⁶⁸

VI. OTHER CONTRIBUTIONS

Walter Gordon's share in the KG equation is based on a paper he sent to *Zeitschrift für Physik* at the end of September.⁶⁹ At that time the KG equation had already been stated by several other physicists. In deriving the KG equation Gordon followed closely the method which Schrödinger had used in June and which also had been used by de Broglie.

The main thing in Gordon's paper was the application of relativistic wave mechanics to the Compton effect. For this purpose he found the current and charge densities by adopting the method used by Schrödinger in his fourth communication. In this paper Schrödinger showed that the quantities

$$\rho = e\psi\psi^*$$

and

$$j_k = \frac{e\hbar}{2mi} \left(\psi^* \frac{\partial\psi}{\partial x_k} - \psi \frac{\partial\psi^*}{\partial x_k} \right)$$

satisfy a continuity equation,

$$\frac{\partial\rho}{\partial t} + \sum \frac{\partial j_k}{\partial x_k} = 0.$$

Schrödinger accordingly interpreted ρ and j_k as (electrical) charge and current densities. Gordon proved that the KG equation requires another charge density which he wrote as

$$\rho = 1/i \left(\psi^* \frac{\partial \psi}{\partial t} - \psi \frac{\partial \psi^*}{\partial t} \right).$$

In the general case of electromagnetic fields present Gordon wrote the charge and current densities as a four-vector,

$$j_\mu = 1/i \left(\psi^* \frac{\partial \psi}{\partial x_\mu} - \psi \frac{\partial \psi^*}{\partial x_\mu} - \frac{2ie}{\hbar c} A_\mu \psi \psi^* \right),$$

where μ runs from 1 to 4 ($x_4 = ict$, $A_4 = i\phi$). These expressions for j_μ were also found by Klein whose paper was independent of Gordon's.⁷⁰

Among the discoverers of the KG equation were, finally, the two Belgian physicists Th. de Donder and Fr. van den Dungen. Their paper was presented in the 5 July 1926 issue of *Comptes Rendus*.⁷¹ De Donder and van den Dungen treated dynamical systems of a very general sort, including gravitation and electromagnetism. Their method was essentially a generalized, covariant version of the variational procedure also applied by Schrödinger and Fock. However, the Belgian physicists did not introduce quantum theory explicitly and did not refer to any works of quantum physicists. It was only in a following note, of October, that de Donder showed that his and van den Dungen's equation could be developed into a relativistic version of the Schrödinger equation.⁷²

De Donder, director of the Institute of Mathematical Physics at the Free University of Brussels, was a very productive scientist. In numerous articles and books he sought to develop an all-embracing, mathematically exact theory of fields and matter. After he and van den Dungen published their equation, he continued to develop it into still more general forms. De Donder's extensive writings were however outside the mainstream of quantum physics and made very little impact on its development. The physicists of de Donder's school, including Rosenfeld, seem to have aimed at grand mathematical schemes rather than physical understanding. As shown by the citation pattern of their many publications, they remained isolated from the trend in quantum physics which took place outside the French-speaking world.

In the fall of 1926 the KG equation was common knowledge among quantum theorists. Apart from the physicists already mentioned, the equation was in 1926 stated also by Paul Dirac in Cambridge and E. Guth in Vienna as well as by other scientists.⁷³ V. Bursian in Leningrad applied the KG equation in order to treat dispersion as a perturbation. Apart from a small relativistic correction he reached the same results as found by Schrödinger by means of the non-relativistic equation.⁷⁴

In the years 1926–1928 the second-order relativistic wave equation (or its equivalent in general quantum mechanics) was considered by many physicists. Most of them treated it in the framework of general relativity, five dimensional or not.⁷⁵ The KG equation did not, however, play an important role in the development of quantum mechanics. Apart from its rather accidental use in the theory of Compton effect and dispersion the equation did not seem effective in solving important physical problems. In particular it failed in incorporating the electron spin in quantum theory.

Apart from its empirical sterility it became difficult to

make sense of the KG equation when considered in the perspective of the general scheme of quantum mechanics such as it became established in 1927 (Born's probabilistic interpretation, Dirac's and Jordan's transformation theory, Heisenberg's uncertainty principle). This, rather than its lack of empirical success, caused some physicists, in particular Pauli and Dirac, to discard it and look for an alternative relativistic equation. In Dirac's search for a new relativistic wave equation he was guided by the insight that in order to comply with general quantum mechanics, the equation has to be linear.⁷⁶ In his fundamental paper on the interpretation of quantum mechanics Heisenberg pointed out the same thing, viz. that a physical understanding of quantum mechanics could not be reached if based on nonlinear equations such as the KG equation.⁷⁷ In a letter to Pauli of February 1927 he mentioned the crucial significance of "the linearity of the transformation—or wave equations," and added that "also in relativistic QM [quantum mechanics] will I at no price give it [the linearity] up."⁷⁸

VII. DEVELOPMENT AFTER 1926

The present work has focused on the origins and very early development of the first relativistic quantum theory. It will be appropriate to include a few words about the role played by the second-order wave equation after 1926.

Although the KG theory virtually disappeared from the scene of physics after Dirac's alternative theory made its entry, it reappeared some years later. In 1934 Pauli and Weisskopf reconsidered the KG equation in the light of current theories of quantization of wave fields.⁷⁹ They emphasized that their new interpretation of the KG equation allowed them to account for the existence of anti-particles in a much more natural way than Dirac's questionable hole theory which was, they said, based on *a priori* arguments. Indeed, Pauli referred to the work as the "anti-Dirac paper."⁸⁰ However, the Pauli–Weisskopf version of the KG equation applies to bosons only and is thus not truly an alternative to Dirac's theory. In 1934 no elementary particles of spin zero were known. It was only later, with the discovery of mesons, that the KG equation proved to apply to reality. In Yukawa's classic theory of 1935 in which the meson (then called the "U-quantum") was predicted, he made use of a version of the KG equation.⁸¹ Since the mid-thirties the KG equation has been recognized to belong to the fundamental equations of quantum field theory.

As mentioned, in the late twenties the five-dimensional theory of Kaluza, Klein, and Fock seemed to be a blind alley. It remained in the periphery of physics for many years, considered to be speculative and of mathematical interest only. However, the five-dimensional approach has experienced a revival too and has recently led to much work in unified field theory. A modern version of the Kaluza–Klein–Fock theory was worked out in the mid-seventies by J. Scherk and others.⁸² The new approach has attracted much interest in connection with the theories of supergravity which were invented at about the same time. As in the original Kaluza–Klein–Fock theory the modern theories operate with more than four dimensions and attribute physical reality to at least some of the extra dimensions. According to one version which is presently investigated by several physicists a unification of the fundamental forces in nature may be obtained in a space-time of eleven dimensions.⁸³ Although the role of Kaluza–Klein–Fock

models is still uncertain, Klein's old theory has proved to contain ideas which are very fruitful to modern physics.

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 - ³H. Kragh, *Centaurus* **26**, 154 (1982). Other historical studies of early wave mechanics include L. Wessels, *Stud. Hist. Phil. Sci.* **10**, 311 (1979); F. Kubli, *Revue d'Histoire des Sciences* **28**, 97 (1975); P. A. Hanle, *Am. J. Phys.* **47**, 644 (1979); P. Forman and V. V. Raman, *Hist. Stud. Phys. Sci.* **1**, 291 (1969); B. Carazza and G. P. Guidetti, *Arch. Hist. Ex. Sci.* **22**, 373 (1980).
 - ⁴L. de Broglie, *Recherches sur la Théorie des Quanta* (Masson, Paris, 1924).
 - ⁵E. Schrödinger, *Ann. Phys.* **79**, 361 (1926), dated 27 January.
 - ⁶E. Schrödinger, *Ann. Phys.* **81**, 109 (1926), dated 21 June.
 - ⁷The expressions in Eq. (7) are squared and then the operators in Eq. (6) are inserted, working on the wave function; to get Eq. (9) one makes use of the Lorentz gauge
- $$\frac{\partial \phi}{\partial t} + c \sum \frac{\partial A_k}{\partial x_k} = 0.$$
- ⁸E. Schrödinger, *Ann. Phys.* **82**, 257 (1927).
 - ⁹Letter from Schrödinger to Weyl, 1 April 1931. Translated from J. U. Gerber, *Arch. Hist. Ex. Sci.* **5**, 349 (1969), on p. 416. It is not entirely clear why Schrödinger wanted to dissociate the KG equation from the name of de Broglie. It may simply have been a matter of priority, but it may also have been rooted in the disagreements between Schrödinger and de Broglie over the interpretation of wave mechanics. In 1931 de Broglie endorsed the Copenhagen interpretation of quantum mechanics to which Schrödinger remained hostile.
 - ¹⁰L. de Broglie, *C. R.* **180**, 498 (1925).
 - ¹¹E. Schrödinger, *Phys. Rev.* **28**, 1049 (1926).
 - ¹²L. de Broglie, in *Wave Mechanics: The First Fifty Years*, edited by W. C. Price *et al.* (Wiley, New York, 1973), p. 15.
 - ¹³L. de Broglie, *C. R.* **183**, 272 (1926), dated 19 July; *J. Phys.* **7**, 321 (1926).
 - ¹⁴Archive for the History of Quantum Physics (AHQP), interview with Klein, 1963; O. Klein, *Sven. Naturvetensk.*, 159 (1973).
 - ¹⁵Reference 14.
 - ¹⁶Reference 14.
 - ¹⁷Reference 14.
 - ¹⁸The present author has in vain looked for relevant sources. A box with Klein's correspondence is deposited at the Bohr Institute, Copenhagen, but contains no letters from the period 1923–1926.
 - ¹⁹Reference 4.
 - ²⁰O. Klein, *Z. Phys.* **37**, 895 (1926), dated 28 April.
 - ²¹Reference 14.
 - ²²Klein's discussions with Pauli are reflected in a letter from Pauli to Jordan of 12 April 1926; Ref. 1, p. 315. Klein also discussed his manuscript with Heisenberg who was then at Bohr's institute; Ref. 20, p. 904.
 - ²³O. Klein, *Sven. Naturvetensk.*, 164 (1973).
 - ²⁴O. Klein, *Z. Phys.* **46**, 190 (1928).
 - ²⁵Reference 20, p. 905; also O. Klein, *Nature* **118**, 516 (1926).
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 - ²⁸Reference 27, p. 30.
 - ²⁹Letter from Einstein to Ehrenfest, quoted from A. Pais, *Subtle is the Lord* (Clarendon, Oxford, 1982), p. 333.
 - ³⁰F. London, *Naturwiss.* **15**, 15 (1927); the idea of interpreting spin effects in terms of five-dimensional relativistic wave mechanics was also explored in L. Rosenfeld, *Bull. Acad. R. Belg.* **13** (5), 326 (1927).
 - ³¹F. London, *Z. Phys.* **42**, 375 (1927).
 - ³²Reference 14.
 - ³³Reference 29, p. 332.
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“Einige Anwendungen der Schwingungsgleichung der Quantentheorie.”

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- ³⁶V. Fock, *Z. Phys.* **39**, 226 (1926), dated 30 July.
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- ⁴⁴Reference 14.
- ⁴⁵L. de Broglie, *J. Phys.* **8**, 244 (1927).
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- ⁴⁸Pauli to Bohr, 6 August 1927; Ref. 1, p. 403.
- ⁴⁹Pauli to Bohr, 29 March 1927; Ref. 1, p. 390. Pauli's comment was to O. Klein, *Z. Phys.* **41**, 407 (1927).
- ⁵⁰Reference 14.
- ⁵¹E.g., A. Schidlof, *C. R.* **185**, 889 (1927); A. Proca, *C. R.* **186**, 739 (1928); D. J. Struik and N. Wiener, *J. Math. Phys.* **7**, 1 (1927); *Proc. Nat. Acad. Sci.* **14**, 262 (1928); F. Gonseth and G. Juvet, *C. R.* **185**, 535 (1927).
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- ⁵³Reference 14.
- ⁵⁴E.g., O. Klein, *Ark. Mat. Astron. Fys.* **34A**, 1 (1948).
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- ⁵⁷Reference 22. Pauli's letter was analyzed by B. L. van der Waerden in *The Physicist's Conception of Nature*, edited by J. Mehra (Reidel, Dordrecht, 1973), p. 276.
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- ⁷⁷W. Heisenberg, *Z. Phys.* **43**, 172 (1927).
- ⁷⁸Heisenberg to Pauli, 23 February 1927; Ref. 1, p. 379.
- ⁷⁹W. Pauli and V. Weisskopf, *Helv. Phys. Acta* **7**, 709 (1934).
- ⁸⁰Cf. V. F. Weisskopf, in *The Birth of Particle Physics*, edited by L. M. Brown and L. Hoddeson (Cambridge University, Cambridge, 1983), p. 70.
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Field Theories of More Than Four Dimensions, edited by V. De Sabbata and E. Schmutzer (World Science Publishers, Singapore, 1983), pp. 427–446. This volume also includes facsimiles and translations of five letters from Einstein to Kaluza written between 1919 and 1925.

Fluctuations in a non-Maxwellian gas

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The velocity distribution of a noble gas in a very strong magnetic field has been shown to be non-Maxwellian. It is assumed here that a torsional suspension is immersed in such a gas. The average potential and kinetic energies of the resultant fluctuation are computed using Langevin's method. They are found to be $(kT/2)$, notwithstanding the reduced population of molecules at higher speeds.

Over a century has passed since the first attempts were made to connect thermodynamics rigorously with the rest of physics via mathematics. But despite much effort that objective remains to be achieved. This is evidenced by the "postulates" on which statistical mechanics is presently based, as well as by the sizable number of papers investigating such antithermodynamic topics as Maxwell Demons, diodes that rectify thermal noise, ferrite isolators that transmit microwave power one way, series motors that produce a unidirectional torque from current fluctuations, and the like. Some of these papers present fairly detailed analyses. These generally show that the hypothetical exception to the second law will not actually occur. Thus van Kampen¹ demonstrated that a diode will not rectify thermal noise. Dreicer² showed that the scrambling of photon energies in a combined photon-electron gas by collisions with the electrons will not alter the Planck distribution of photon energies (any change in which would constitute a violation of the second law). McFee³ has identified the fallacies in several schemes for circumventing the second law through the use of fluctuations in the effective temperature of small particles coupled to a heat reservoir. These studies are useful in that they help to increase our understanding of the origins of the universality and exactness of thermal physics, in much the same way that the various tests of special relativity⁴ helped to convince physicists of the general validity of that theory.

In this paper we consider a new hypothetical thermodynamic anomaly. It involves an exotic situation deliberately contrived to smoke the second law, so to speak, out of its lair. We study the fluctuations of a torsional suspension (such as a galvanometer) that is immersed in a noble gas (e.g., neon) which is subjected to a magnetic field as strong as those found near neutron stars. Such fields are of the order of 10^7 tesla,⁵ i.e., about one million times stronger than the most intense steady field we can now produce in the laboratory. Obviously we are concerned with a conceptual rather than practical situation. For reasons that will be clear shortly, the gas is called a "magneto-electric polarization" (MEP) gas. It has been shown⁶ that in this gas there

are fewer atoms moving at higher thermal speeds than there are in an ordinary Maxwellian gas. We are interested in the amount of jiggle produced in the torsional suspension by the bombardment of this gas. Since it is the high-speed particles that are responsible for the lion's share of the jiggle, it would seem that the fluctuations of a suspension immersed in the MEP gas would be smaller than those of one immersed in an ordinary gas. If so, this would conflict with the second law.

The equations of motion⁷ for the particles of the MEP gas, and the velocity distribution⁶ derived from them, have been discussed in two prior papers. Before plunging into the calculation of the fluctuation, these papers will be reviewed briefly.

The gas is assumed to be at a modest temperature, e.g., 300 K. As a result essentially all of the atoms are in their ground state. The magnetic field alters the orbits of the atom's electrons, changing the shape of the atom and inducing a (diamagnetic) magnetic moment which has an associated magnetic polarization energy. The atom moves about thermally at a speed of the order of 1000 m/s. Moving cross field at this speed it sees⁸ an electric field $b(V \times B)$ where (b) is the relativistic factor $[1 - (w/c)^2]^{-1/2}$, (w) the cross-field velocity, and (c) the speed of light. The magnetic field is also modified, becoming (bB) . At 1000 m/s (b) differs from 1 by only a tiny amount, about $(10^{-11}/2)$. Thus the magnetic energy stays essentially the same, and the electric field can be taken to be $(V \times B)$. When the cross-field component of (V) is 1000 and B is 10^7 T this electric field is enormous, 10^{10} V/m. It is not much less than that required to rip an electron away from the atom thus ionizing it. A large dipole moment is produced, having an energy of electric polarization so large that it is a substantial fraction of the kinetic energy. At "low" speeds this polarization energy varies as the square of speed as does the kinetic energy and thus increases the effective mass. At high thermal speeds the electric field becomes so strong that the induced electrical polarization exhibits partial saturation. That such saturation occurs has been shown experimentally using high-power lasers.⁹ To simplify calcula-